

WEEK 2 PORTFOLIO PROJECT: ENTERPRISE INFRASTRUCTURE
PLANNING FOR A STARTUP COMPANY

PART 1 – COMPANY PROFILE

Company Name: ABC Startup Solutions

Nature of Business: Software Development

Company Vision

To become a trusted software development company that creates reliable and innovative software solutions for businesses.

Company Mission

ABC Startup Solutions is committed to developing reliable, innovative, and user-focused software solutions that address the needs of businesses, improve their operations, and support their growth through the effective use of technology.

Office Location: Bonifacio Global City, Taguig City, Philippines

Organizational Structure



Employee Distribution

DEPARTMENT	NO. OF EMPLOYEES
Information Technology	5
Human Resources	4

Finance	5
Sales	6

PART 2 – ENTERPRISE HARDWARE INVENTORY

ASSET ID	HARDWARE	QUANTITY	DEPARTMENT	PURPOSE
HW-001	Desktop Computer	20	IT (5), HR (4), Finance (5), Sales (6)	Provides each employee with a workstation for daily business tasks.
HW-002	Laptop	5	IT	Provides IT employees with portable computers for technical work and troubleshooting.
HW-003	Server	1	IT	Provides centralized IT services and supports company data and applications.
HW-004	Router	1	All Departments	Provides internet connectivity and manages network traffic.
HW-005	Switch	1	All Departments	Connects wired computers and network devices within the office.
HW-006	Printer	2	All Departments	Provides shared printing for business documents and records.
HW-007	UPS	6	IT/Critical Equipment	Provides temporary backup power and protection for important IT equipment.

HW-008	Wireless Access Point	2	All Departments	Provides wireless network access throughout the office.
HW-009	NAS Storage	1	IT	Provides centralized network storage for company files and data.
HW-010	External Backup Drive	2	IT	Provides centralized network storage for company files and data.
HW-011	Monitor	20	IT (5), HR (4), Finance (5), Sales (6)	Provides a display for each desktop computer.

Why these quantities?

- **20 Desktops Computers** – One for each employee. Because each employee needs a workstation to perform their daily tasks.
- **5 Laptops** – One for each IT employee. We don't necessarily need 20 laptops because everyone already has a desktop. We can assign laptops primarily to the IT department for tasks that may require mobility, such as troubleshooting, infrastructure maintenance, and working away from a fixed workstation.
- **1 Server** – The server is part of the company's central IT infrastructure. IT will manage it, but it can provide services to the entire company.
- **1 Router** – The router provides network/internet connectivity for the whole office.
- **1 Switch** – The switch connects the wired computers and other network devices. So all departments use the network provided by the switch.
- **2 Printers** – Shared by the employees instead of purchasing one printer per person. Two printers also provide another option if one is unavailable.
- **6 UPS** – We can provide UPS units for the important equipment rather than every single computer.
- **2 Wireless Access Point** – Provides wireless coverage for the single office floor.
- **1 NAS Storage** – It provides centralized storage that can be accessed through the company network. IT manages it, while authorized employees/departments can use the stored files.
- **2 External Backup Drives** – Having two gives the company additional backup capacity and redundancy.
- **20 Monitors** – We have 20 desktop computers, so we need 20 monitors.

PART 3 – ENTERPRISE SOFTWARE INVENTORY

SOFTWARE	VERSION	LICENSE	PURPOSE
Windows 11 Pro	25H2	Commercial	Provides the operating system for employee desktop and laptop computers.
Ubuntu Server	24.04 LTS	Open Source	Provides the operating system for the company's server.
Microsoft Office	Microsoft 365 Apps	Subscription	Provides applications for creating documents, spreadsheets, presentations, and other office work.
VS Code	1.103	Free	Provides a development environment for employees working on software development tasks.
Git	2.50.1	Open Source	Provides version control for managing and tracking changes in source code.
GitHub Desktop	3.5.2	Free	Provides a graphical interface for managing Git repositories and software development projects.
VirtualBox	7.2.14	Open Source	Allows IT staff and developers to create and manage virtual machines for testing and development.
Google Chrome	139.0	Free	Provides a web browser for accessing websites, web applications, and online services.
Microsoft Defender	4.18	Included with	Provides malware

		Windows	and security protection for Windows computers.
AnyDesk	9.0	Commercial / Free Options	Provides remote access and support for troubleshooting computers and assisting users.
7-Zip	26.02	Open Source	Provides file compression and extraction for managing compressed files.

Why each software is needed?

1. Windows 11 Pro – provides the operating system needed for the company's desktop computers and laptops. It allows employees to run business applications and perform their daily tasks securely.

2. Ubuntu Server – provides the operating system for the company's server. It can support centralized services, file management, and other server-based operations required by the company.

3. Microsoft Office – provides productivity applications for creating documents, spreadsheets, presentations, and other business files. It is useful for departments such as HR, Finance, Sales, and IT in performing their daily office tasks.

4. Visual Studio Code – provides developers with a tool for writing and editing source code. It is important for the IT department because ABC Startup Solutions is a software development company.

5. Git – is used to track changes made to source code during software development. It helps developers manage different versions of their projects and work together more efficiently.

6. GitHub Desktop – provides a graphical interface for managing Git repositories. It makes it easier for developers to commit changes, manage branches, and synchronize their software projects.

7. VirtualBox – allows IT employees to create and run virtual machines on their computers. It can be used to test software and different operating systems without affecting the main computer environment.

8. Google Chrome – provides a web browser that employees can use to access websites, web applications, and online services. It is useful for daily business activities such as research, communication, and accessing cloud-based systems.

9. Microsoft Defender – provides antivirus and malware protection for computers running Windows. It helps protect company devices and data from common security threats.

10. AnyDesk – allows authorized IT personnel to remotely access computers for technical support and troubleshooting. This can help the IT department resolve certain computer problems without needing to physically access the employee's workstation.

11. 7-Zip – is used to compress and extract files in different archive formats. It helps employees manage large files and access compressed files received from other users or systems.

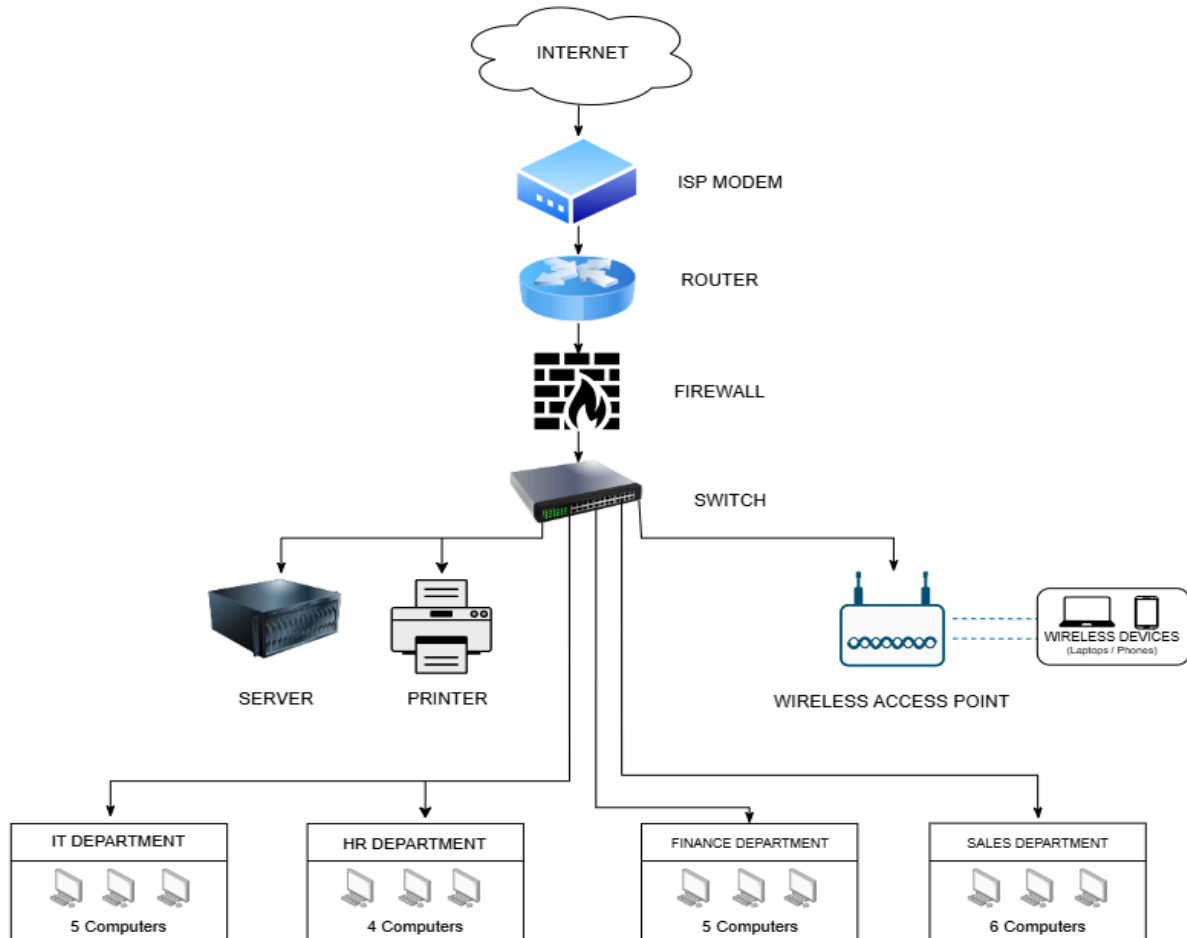
PART 4 – ENTERPRISE NETWORK INVENTORY

ASSET ID	NETWORK EQUIPMENT	QUANTITY	PURPOSE
NET-001	ISP Modem	1	Connects the company network to the Internet Service Provider.
NET-002	Router	1	Routes network traffic between the internal network and the internet.
NET-003	Firewall	1	Filters network traffic and helps protect the company network from unauthorized access.
NET-004	Switch	1	Connects wired computers, servers, printers, and other network devices.
NET-005	Access Point	2	Provides wireless network connectivity throughout the office.
NET-006	Patch Panel	1	Organizes and manages network cables in the office.
NET-007	CAT6 Cables	1 Box	Provides wired network connections between computers and network equipment.
NET-008	RJ45 Connectors	1 Pack	Provides connectors for terminating CAT6 network cables.

PART 5 – ENTERPRISE NETWORK DIAGRAM

ABC STARTUP SOLUTIONS ENTERPRISE NETWORK TOPOLOGY

Prepared by: Shiella Mae B. Rico



PART 6 – SYSTEM ADMINISTRATION ROLES

Helpdesk Technician

- **Responsibilities:** Provides technical assistance to users, diagnoses computer problems, sets up or repairs hardware and software, and documents user issues. Helpdesk technicians may also install software and assist users with hardware and operating-system problems.
- **Skills:** Troubleshooting, problem-solving, communication, customer service, hardware and software knowledge, and basic networking.
- **Common Tools:** Helpdesk/ticketing systems, remote-support software, Windows diagnostic tools, hardware diagnostic tools
- **Certifications:** CompTIA A+ , CompTIA Security+

Network Administrator

- **Responsibilities:** Installs, configures, maintains, monitors, and troubleshoots computer networks. They also maintain network equipment such as routers and switches and help keep the network secure.
- **Skills:** Networking, TCP/IP, network configuration, troubleshooting, network security, routing, and switching.
- **Common Tools:** Router and switch management tools, network monitoring tools, Wireshark, ping, traceroute
- **Certifications:** Cisco Certified Network Associate (CCNA), Microsoft Azure Network Engineer Associate, CompTIA Network+

Linux System Administrator

- **Responsibilities:** Manages Linux systems and servers, including software, services, storage, users, networking, permissions, and security. System administrators also maintain and troubleshoot Linux environments.
- **Skills:** Linux command line, system administration, shell scripting, storage management, networking, user management, and security.
- **Common Tools:** Linux Terminal, SSH, systemd, shell scripts, package managers, firewall tools
- **Certifications:** Red Hat RHCSA, Linux Foundation Certified System Administrator (LFCS), CompTIA Linux+

Cloud Administrator

- **Responsibilities:** Implements, manages, and monitors cloud resources including identity, storage, computing, networking, security, and governance. Cloud administrators also monitor resources and manage backup and recovery operations.
- **Skills:** Cloud computing, networking, virtualization, storage, identity management, security, monitoring, and backup/recovery.
- **Common Tools:** Microsoft Azure Portal, Azure CLI, PowerShell, Azure Monitor, Azure Storage tools
- **Certifications:** Azure Administrator Associate (AZ-104), AWS Certified CloudOps Engineer – Associate, AWS Certified Cloud Practitioner, Google Cloud Associate Cloud Engineer

How these four professionals work together?

The Helpdesk Technician, Network Administrator, Linux System Administrator, and Cloud Administrator work together to maintain the organization's IT infrastructure. The Helpdesk Technician receives and troubleshoots user issues and escalates problems that require specialized assistance. The Network Administrator handles network connectivity and equipment, while the Linux System Administrator manages Linux based servers and services. The Cloud Administrator manages cloud resources, security, storage, and computing services, allowing the four professionals to coordinate when problems involve multiple areas of the organization's IT environment.

References:

- U.S. Bureau of Labor Statistics (BLS) — Computer Support Specialists
<https://www.bls.gov/ooh/computer-and-information-technology/computer-support-specialists.htm>

- Cisco — Network Administrator
<https://www.cisco.com/site/us/en/learn/training-certifications/tech-roles/network-administrator.html>
- Red Hat — RHCSA
https://www.redhat.com/en/services/certification/rhcsa?utm_source=chatgpt.com
- Red Hat — RHCSA Exam Objectives
<https://www.redhat.com/en/services/training/ex200-red-hat-certified-system-administrator-rhcsa-exam>
- Linux Foundation — LFCS <https://training.linuxfoundation.org/certification/LFCS>
- Microsoft Learn — AZ-104 Study Guide
<https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/az-104>

PART 7 – INFRASTRUCTURE RECOMMENDATIONS

1. Internet Provider

- **Recommendation:** PLDT Enterprise — Affordaboost 500 Mbps
 - **Internet Speed:** Up to 500 Mbps
 - **Connection Type:** Fiber
 - **Static IP:** Included
 - **Modem:** Wi-Fi modem included
 - **Contract:** 24-month term
 - **Monthly Plan:** ₱2,000/month
 - **Provider:** PLDT Enterprise
- **Logical Justification:** A 500 Mbps business fiber connection is appropriate for a 20-employee software development company because employees will need internet access for software development, cloud services, video meetings, file transfers, online collaboration, and business communication. PLDT Enterprise currently lists its Affordaboost 500 Mbps plan with up to 500 Mbps bandwidth, a Wi-Fi modem, and a free static IP, making it a practical option for a small office.

2. Server Specifications & Environmental Considerations

- **Recommendation:** 1U Rackmount Server — Dell PowerEdge R360
 - **Processor:** Intel Xeon E-2400 series, up to 8 cores
 - **RAM:** 32 GB DDR5 ECC memory
 - **Storage:** 4 × 2 TB enterprise SSDs
 - **RAID:** RAID 10
 - **Operating System:** Ubuntu Server 24.04 LTS
 - **Power Supply:** Redundant hot-swappable power supply
 - **UPS:** Heavy-duty UPS for protection against power interruptions
 - **Cooling:** Rack-mounted air cooling with adequate ventilation
- **Logical Justification:** The Dell PowerEdge R360 is a 1U server that supports an Intel Xeon E-2400 series processor with up to eight cores and ECC DDR5 memory, making it suitable for a small organization's server environment. Its storage-controller options and support for multiple SAS/SATA drives also provide flexibility for a redundant RAID configuration. The server should also be connected to a UPS because an unexpected

power interruption can interrupt services and potentially cause data corruption or hardware problems.

3. Backup Strategy

- **Recommendation:** Implementation of the 3-2-1 Backup Rule
 - **3 Copies of Data:** One primary production copy, one local backup copy, and one off-site backup copy.
 - **2 Different Media Types:** NAS storage for local backups and external hard drives for offline backups.
 - **1 Copy Stored Off-site:** Encrypted cloud storage should be used for an additional backup copy.
 - **Backup Schedule:** Daily backups for important company data, with regular verification and restoration testing.
 - **Offline Backup:** At least one backup should be disconnected from the network when not being used.
- **Logical Justification:** The 3-2-1 strategy reduces the risk of losing all copies of important company data because the backups are stored in different locations and media. CISA recommends businesses back up important data, while its cybersecurity guidance also emphasizes maintaining and testing backups; offline or physically separate backups provide additional protection against ransomware.

4. Security Recommendations

- **Recommendation:** Implement basic security controls before the company's network becomes operational.
 - **Firewall:** Deploy a dedicated firewall between the router and internal network.
 - **Network Segmentation:** Separate important network resources and provide a separate guest Wi-Fi network.
 - **Access Control:** Give employees access only to the systems and files required for their job.
 - **Software Updates:** Keep operating systems, applications, servers, and network equipment updated.
 - **Multi-Factor Authentication:** Enable MFA for email, cloud services, administrator accounts, and remote access.
 - **Data Protection:** Protect sensitive company and employee information through access controls and encryption where appropriate.
 - **Security Monitoring:** Monitor important network and server activity and maintain logs.
- **Logical Justification:** The company currently has no security policies or infrastructure, so security controls should be established before employees begin using the network. CISA's guidance for small and medium-sized businesses emphasizes MFA, software updates, backups, encryption, logging, and strong passwords as important cybersecurity practices.

5. Antivirus

- **Recommendation:** Microsoft Defender Antivirus
 - Enable Real-time Protection on all Windows 11 Pro computers.

- Keep security intelligence updates enabled.
- Enable cloud-delivered protection.
- Enable Tamper Protection where supported.
- Perform regular antivirus scans.
- Configure Controlled Folder Access for important files where appropriate.
- Avoid running multiple real-time antivirus programs simultaneously.
- **Logical Justification:** Microsoft Defender Antivirus is already built into Windows 11 and provides real-time, behavior-based, and cloud-delivered protection against malware and other threats. Using the built-in protection also reduces unnecessary software complexity and additional licensing costs for the startup.

6. Password Policy

- **Recommendation:** Establish a company-wide password and authentication policy.
 - **Minimum Length:** Require passwords to be at least 15 characters when passwords are used as a single authentication factor.
 - **Passphrases:** Encourage employees to use long and memorable passphrases.
 - **Unique Passwords:** Employees must use different passwords for different accounts.
 - **Compromised Passwords:** Block commonly used or compromised passwords.
 - **Password Manager:** Allow the use of an approved password manager.
 - **MFA:** Require MFA for administrative accounts, email, cloud services, and remote access.
 - **Password Changes:** Do not require routine password changes unless there is evidence of compromise or another legitimate security reason.
 - **Account Lockout/Rate Limiting:** Limit repeated failed login attempts.
- **Logical Justification:** The password policy follows current NIST guidance, which emphasizes password length, blocking commonly used or compromised passwords, allowing password managers, and avoiding unnecessary periodic password changes. NIST's current guidance specifies a minimum of 15 characters for passwords used as a single-factor authentication mechanism and recommends MFA for stronger authentication.

7. Expansion Plan

- **Recommendation:** Design the infrastructure so that ABC Startup Solutions can expand without replacing its entire network.
 - **Network Ports:** Use switches with additional available ports for future employees and devices.
 - **Wi-Fi:** Add additional access points if office coverage or employee numbers increase.
 - **Server:** Use a server platform with expandable RAM and storage capacity.
 - **Storage:** Expand the NAS and backup storage as company data increases.
 - **Internet:** Upgrade the fiber plan if bandwidth requirements increase.
 - **Cloud Services:** Gradually adopt cloud services for additional storage, backup, development, and business applications.
 - **Future Employees:** Reserve network addresses, switch ports, workstation space, and equipment capacity for new employees.

- **Additional Office Locations:** If the company opens another branch, consider VPN or SD-WAN connectivity between locations.
- **Logical Justification:** Planning for expansion prevents the company from having to replace its entire infrastructure when the business grows. The recommended server and network equipment should therefore have available capacity so that additional users, storage, network devices, and services can be added as needed.

PART 8 – PERSONAL REFLECTION

Through this project, I learned that system administration involves more than simply setting up computers and installing software. I learned how to analyze the needs of a company and use those requirements to plan its hardware, software, network infrastructure, security, server, backup strategy, and other IT resources. I also learned how different departments have different technology needs and how these needs must be considered when designing an infrastructure. Creating the network diagram also helped me understand how devices such as routers, firewalls, switches, servers, and access points work together in an organization.

The most challenging task for me was designing the enterprise network diagram and deciding how the different network devices should be connected. I had to understand the purpose of each device and make sure that the connections were logical and appropriate for the company's requirements. Organizing all the information needed for the project and making sure that each recommendation matched the company's needs was also challenging because I had to carefully consider the appropriate hardware, software, network setup, security measures, and backup strategies to create a practical IT infrastructure.

Planning is important before deployment because it helps prevent unnecessary expenses, configuration problems, security risks, and compatibility issues. A System Administrator needs to understand the organization's requirements before purchasing equipment or implementing services. Proper planning also makes it easier to estimate costs, prepare for future expansion, establish security measures, and ensure that the infrastructure can support the company's operations. Without proper planning, an organization may purchase unsuitable equipment or create an infrastructure that cannot meet its actual needs.

This project will help me become a better System Administrator by improving my ability to analyze requirements, make technical decisions, document infrastructure, and solve problems. It also gave me experience in planning an IT environment from the beginning instead of focusing only on individual devices or software. Most importantly, I learned that a System Administrator must consider performance, security, reliability, scalability, and business requirements when designing an IT infrastructure. This experience will help me become more prepared for future system administration tasks and real-world IT environments.